

Alignment of Correlated Random Graphs

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Motivation

Erdős–Rényi alignment

Power law alignment

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Power law alignment

Graph alignment problem

- G : random graph with vertex set $[n]$
- G_1, G_2 : two independent edge-subsamplings of G
 - Keep each edge with probability s
- Sample a permutation $\pi^* \in S_n$ uniformly at random
- Permute the vertex labels of G_2 by π^* to obtain G'_2
- **Problem:** given G_1 and G'_2 , can you find π^* ?

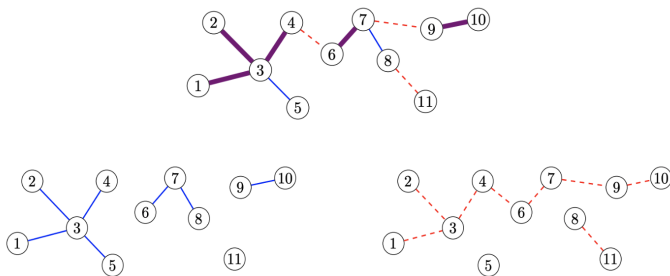


Figure: Figure 1 of [GML21a]. A realization of (G, G_1, G_2) .

Notions of recovery

Let $\hat{\pi} : \mathcal{C} \rightarrow [n]$ be a proposed (partial) matching.

Overlap and error fraction of $\hat{\pi}$, respectively:

$$\text{ov}(\pi^*, \hat{\pi}) = \frac{1}{n} \sum_{u \in \mathcal{C}} \mathbf{1}_{\hat{\pi}(u) = \pi^*(u)}$$

$$\text{err}(\pi^*, \hat{\pi}) = \frac{1}{n} \sum_{u \in \mathcal{C}} \mathbf{1}_{\hat{\pi}(u) \neq \pi^*(u)}$$

- Exact recovery: $\hat{\pi}(u) = \pi^*(u)$ for every $u \in [n]$ whp
- Partial recovery: there exists a constant $\eta > 0$ such that $\text{ov}(\pi^*, \hat{\pi}) > \eta$ whp
- One-sided partial recovery: $\text{partial} + \text{err}(\pi^*, \hat{\pi}) = o(1)$ whp

Applications

Protein-protein interaction (PPI) networks

- Proteomics: the study of the collection of proteins in a cell
- Matching PPI networks reveals common biological pathways: [SXB07], [BGSW13]

Deanonimization

- Pioneering works: [NS08], [NS09]
- Netflix Prize dataset: can deanonymize with IMDb ratings
- Social networks: can deanonymize even with small overlap

Real world networks

- Erdős–Rényi: well-understood homogeneous model
- Real world networks: highly inhomogeneous [BB03]
- Real world power law: $P(\text{degree} = k) \propto k^{-\beta}$ for $\beta \in (2, 3)$
 - High-degree “hub” vertices
- This talk: both Erdős–Rényi + power law models

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Literature review

Partial alignment in sparse regime: $G \sim ER(n, \frac{\lambda/s}{n})$

- [DD23], [GML21b]: information-theoretically feasible if $\lambda s > 1$
- [GM20]: poly-time one-sided partial recovery if $\lambda \in (1, \lambda_0]$ and s is sufficiently close to 1
 - Neighborhood Tree Matching Algorithm
- [GML21a], [GMS22]: poly-time one-sided partial recovery if $s > \sqrt{\alpha}$ and λ is sufficiently large
 - Otter's constant: $\alpha \approx 0.338$
 - Correlation detection using a likelihood ratio test
- [MWXY23]: poly-time partial recovery if $s > \sqrt{\alpha}$ and λ is sufficiently large

Conjectured phase diagram

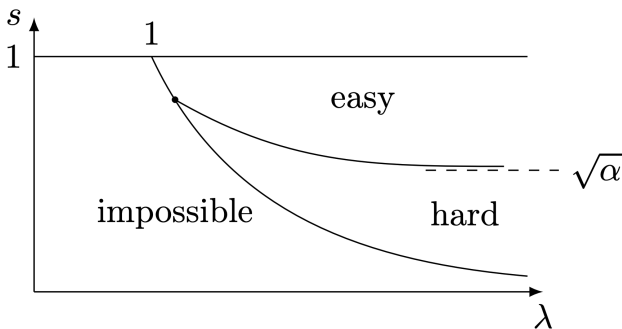


Figure: Figure 2 of [GMS22]. Conjectured phase diagram of sparse Erdős–Rényi partial alignment.

Neighborhood Tree Matching Algorithm

Neighborhood Tree Matching Algorithm (NTMA) of [GM20]

- Reduce to hypothesis testing on pairs of local neighborhoods
- **Idea:** $(v, u) \in V(G_1) \times V(G_2)$ should be matched if their local neighborhoods have large overlap
- In sparse Erdős–Rényi case, local neighborhoods are tree-like
 - For a correct match (v, v) , $(T, T') \sim GW(\lambda, s)$
 - If (v, u) are far apart, $(T, T') \stackrel{iid}{\sim} GW(\lambda)$
- Which metric on trees should we use?

Matching weight

- Let T, T' be trees rooted at ρ, ρ'
- The **matching weight** $M_d(T, T')$ is the largest number of leaves of depth d in any common subtree of T, T'

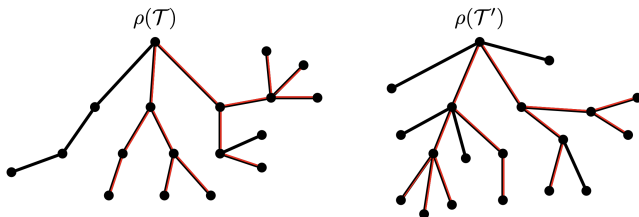


Figure: Figure 1 of [GM20]. A pair of trees with $W_3(T, T') = 7$.

Matching rate

If T, T' are random, how to quantify the matching weight?

- The **matching rate** $\gamma(T, T')$ is the infimum of $\gamma > 0$ such that the tails of $\gamma^{-d} M_d(T, T')$ are subexponential for $d \geq d_0$

Results stated in terms of the matching rate:

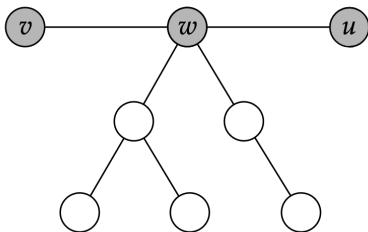
- **Thm 1:** if $(T, T') \sim GW(\lambda, s)$, then $\gamma(\lambda, s) \geq \lambda s$
- **Thm 2:** if $(T, T') \stackrel{iid}{\sim} GW(\lambda)$, then $\gamma(\lambda) < \lambda$ for $\lambda \in (1, \lambda_0]$

Naive version of NTMA

Naive implementation of NTMA:

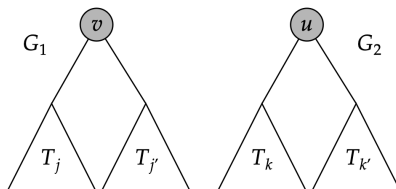
- Given $\lambda \in (1, \lambda_0]$, suppose s satisfies $\lambda s > \gamma(\lambda)$
- Then $\gamma(\text{correctly matched pair}) > \gamma(\text{incorrectly matched pair})$
- Rule: match $(v, u) \in V(G_1) \times V(G_2)$ if $W_d(T_v, T'_u) \geq (\lambda s)^d$

Issue: a common neighbor leads to a high matching weight!



NTMA

Fix: compare a pair of dangling subtrees.



Correct formulation of NTMA:

- Fix a threshold γ and a depth d
- For each $(v, u) \in V(G_1) \times V(G_2)$, if depth d neighborhoods are both trees, and if $W_d(T_j, T_k), W_d(T_{j'}, T_{k'}) > \gamma^{d-1}$, then match (v, u)

Choose d to ensure neighborhoods are trees whp, and choose γ based on **Thm 1** + **Thm 2**.

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Literature review

Seeded algorithms:

- Seeded: a small subset of the matching is revealed
- Percolation methods: include a new edge if it is “close” to many previously-matched edges
 - Simplest: (v, u) and (v', u') are *1-hop neighbors* if $(v, v') \in E(G_1)$ and $(u, u') \in E(G_2)$
- [CGL16]:
 - Chung-Lu model with $\beta \in (2, 3)$
 - Partition the graph into slices of similar degree
 - Count 1-hop neighbors among previously-matched pairs
 - Requires n^ϵ seeds for exact recovery if one can choose them

Literature review

Seedless algorithms:

- [DMWX21]:
 - Erdős–Rényi model
 - “Degree profile matching”: for each vertex v , count how many neighbors have a given outdegrees, and construct a vector of counts. Compare two vertices using the ℓ^1 distance between their vectors
 - Exact recovery if $s = 1 - O((\log n)^{-2})$, min degree $O((\log n)^2)$
- [DFW23]:
 - Generalize [DMWX21] to inhomogeneous Erdős–Rényi model
 - Continuous version of degree profiles. Compare two vertices using the L^1 distance between these functions
 - Exact recovery if $s = 1 - O((\log n)^{-2})$, min degree $O((\log n)^2)$

This talk: seedless partial matching in a simplified power law model at constant s , combining degree profiles + percolation.

Stratified power law model

Vertex set:

- Vertices separated into layers L_k for $0 \leq k \leq k_{\max}$
- $k_{\max} = \lfloor \frac{1}{\beta-1} \log_2 n \rfloor$
- $|L_k| = \lceil C_1 2^{-(\beta-1)k} n \rceil$ for $1 \leq k \leq k_{\max}$

Edge probabilities:

- Assign the weight $w_v = 2^k$ to each vertex $v \in L_k$
- w_{tot} = total sum of weights
- Connect each edge $(v, u) \in L_j \times L_\ell$ independently with probability

$$p_{\ell j} = 1 \wedge \frac{w_v w_u}{w_{\text{tot}}} = 1 \wedge C_2 \frac{2^\ell 2^j}{n}$$

- For $\ell \leq (\beta - 2)k_{\max} - \omega(1)$, we simply have $p_{\ell j} = C_2 \frac{2^\ell 2^j}{n}$

Stratified power law model

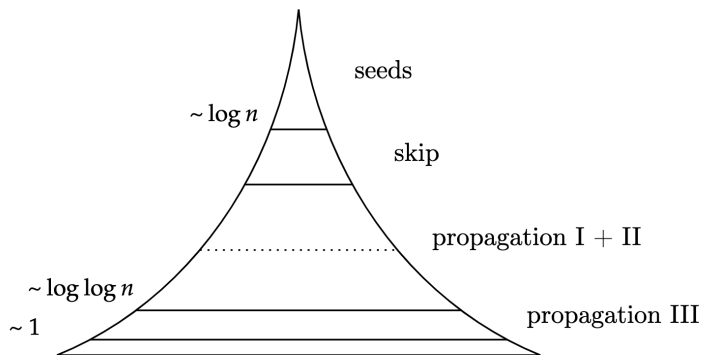
Restating the graph alignment problem:

- G : power law graph described above, with vertex set $[n]$
- G_1, G_2 : two independent edge-subsamplings of G
 - Keep each edge with constant probability $s \in (0, 1]$
- Sample a permutation $\pi^* \in S_n$ uniformly at random
- Permute the vertex labels of G_2 by π^* to obtain G'_2
- *Reveal* the sets $L_k^{(1)} \subseteq V(G_1)$ and $L_k^{(2)} \subseteq V(G'_2)$ (!)
- **Problem:** given G_1 and G'_2 , can you partially recover π^* ?

Result: one-sided partial recovery for constant $s \in (0, 1]$.

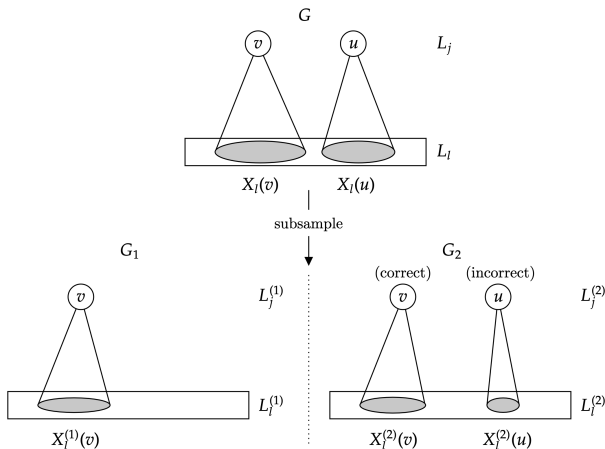
Algorithm: first find seeds, then apply three stages of propagation.

Schematic



Finding seeds

Idea: the correct match $(v, v) \in L_j^{(1)} \times L_j^{(2)}$ has similarly-sized neighborhoods in layer L_ℓ , compared to an incorrect match (v, u) .
 $\leadsto$ Consider an average of $|X_\ell^{(1)}(v) - X_\ell^{(2)}(u)|$ over some set $\ell \in \mathcal{I}$.



Finding seeds

Formalizing our version of degree profile matching:

- $X_\ell^{(1)}(v)$ = number of neighbors of v in $L_\ell^{(1)}$
- $X_\ell^{(2)}(u)$ = number of neighbors of u in $L_\ell^{(2)}$
- Define normalized degree-difference

$$Z_\ell(v, u) = \frac{1}{\sqrt{N_\ell p_{\ell j}}} (X_\ell^{(1)}(v) - X_\ell^{(2)}(u))$$

- Let $\mathcal{I}$ denote the lowest $(1 - o(1))(\beta - 2)k_{\max}$ layers
- Define the mean absolute deviation statistic

$$T(v, u) = \frac{1}{|\mathcal{I}|} \sum_{\ell \in \mathcal{I}} |Z_\ell(v, u)|$$

- Rule: match $v \in L_j^{(1)}$ to vertex $u \in L_j^{(2)}$ with minimal $T(v, u)$
- Repeat for all j among the top $\frac{\alpha}{\beta-1} \log_2 n - o(1)$ layers

Finding seeds

Let $\alpha = 0.009 \frac{\beta-2}{\beta-1} (1 - \sqrt{1-s})^2$.

Claim: we correctly match the n^α highest-degree vertices whp.

- $Z_\ell(v, v) \stackrel{d}{\approx} N(0, 2s(1-s))$, so $\frac{1}{\sqrt{s}} ET(v, v) \sim \sqrt{1-s} \frac{2}{\sqrt{\pi}}$
 - $Z_\ell = \frac{1}{\sqrt{Np}}$ (sum of N rvs with mean 0, variance $2ps(1-s)$)
- $Z_\ell(v, u) \stackrel{d}{\approx} N(0, 2s)$ for $u \neq v$, so $\frac{1}{\sqrt{s}} ET(v, u) \sim \frac{2}{\sqrt{\pi}}$
 - $Z_\ell = \frac{1}{\sqrt{Np}}$ (sum of N rvs with mean 0, variance $2ps(1-ps)$)
- Subexponential tail bound in either case:

$$P\left(\frac{1}{\sqrt{s}} |T(v, u) - ET(v, u)| \geq x\right) \leq 2 \exp\left(-\frac{1}{2} |x| \theta_1 x\right)$$

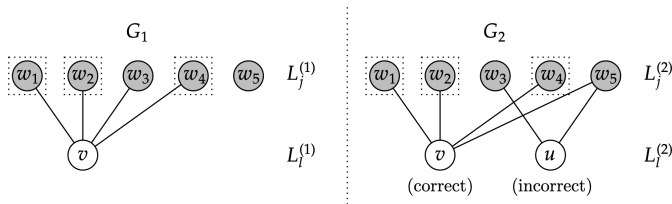
- Set $x = (1 - \varepsilon) \cdot \frac{1}{2} \frac{2}{\sqrt{\pi}} (1 - \sqrt{1-s})$ to find $P(\text{error}) \leq n^{-\gamma}$

Propagation to log log height

How to use a previously-matched layer L_j to match a new layer L_ℓ , with $\ell < j$?

Idea: the correct match $(v, v) \in L_\ell^{(1)} \times L_\ell^{(2)}$ has more 1-hop neighbors, compared to any incorrect match (v, u) .

Rule: match $v \in L_\ell^{(1)}$ to vertex $u \in L_\ell^{(2)}$ with $\max A_j(v, u)$, where $A_j(v, u)$ = number of 1-hop neighbors in layer j .



Propagation to log log height

Claim: if $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$ with $u \neq v$, if layer j was previously-matched, if $N_j p_{\ell j} \rightarrow \infty$, and if $p_{\ell j} \rightarrow 0$, then $P(A_j(v, v) \leq A_j(v, u)) \leq 3 \exp(-\frac{1}{3} N_j p_{\ell j} s^2)$.

- $A_j(v, v) \sim \text{Bin}(N_j, p_{\ell j} s^2)$
- $A_j(v, u) \sim \text{Bin}(N_j, p_{\ell j}^2 s^2)$ (a much smaller mean)
- Apply Chernoff to each binomial to finish

Define:

- $j_{\min} = k_{\max} - \zeta \log_2 n + o(1)$: the lowest seed layer
- $\ell_{\max} = (\beta - 2)k_{\max} - o(1)$
- $\ell_{\text{mid}} = (\beta - 2)k_{\max} - (\beta - 2)\zeta \log_2 n + o(1)$
- $\ell_{\min} = \frac{c_1}{3-\beta} \log_2 \log_2 n$ for $c_1 > 1$

Propagation I: use $L_{j_{\min}}$ to match L_ℓ for $\ell \in [\ell_{\max}, \ell_{\text{mid}}]$ whp.

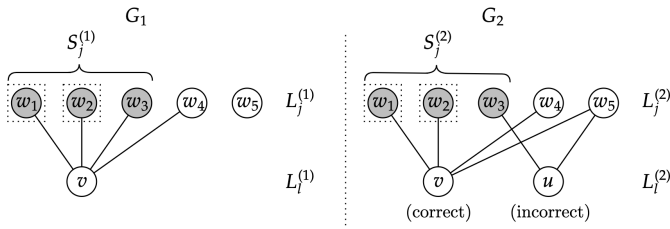
Propagation II: use $L_{\ell+1}$ to match L_ℓ for $\ell \in [\ell_{\text{mid}} - 1, \ell_{\min}]$ whp.

Propagation to constant height

We have reached layer $\frac{c_1}{3-\beta} \log_2 \log_2 n$. We will begin to encounter isolated vertices. Thus we resort to *partially* matching layers.

Idea: same as before, but only consider 1-hop neighbors among the (proper) subset of matched vertices.

Rule: match $v \in L_\ell^{(1)}$ to vertex $u \in L_\ell^{(2)}$ with $\max \tilde{A}_j(v, u)$, **only if** $\tilde{A}_j(v, u) \geq 3$, where $\tilde{A}_j(v, u)$ = number of 1-hop neighbors among partial matching in layer $j = \ell + 1$.



Propagation to constant height

Claim: let $g(x) = 1 - e^{-x}(1 + x + \frac{x^2}{2})$. If $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$ with $u \neq v$, if $M_j = \lambda_j N_j$ vertices have been matched in layer $j = \ell + 1 \leq j_{min}$, then $P(\tilde{A}_j(v, u) \geq 3) = O(\frac{1}{n^{3-o(1)}})$ and $P(\tilde{A}_j(v, v) \geq 3) \geq (1 - \frac{1}{n^{1-o(1)}})g(M_j p_{\ell j} s^2)$.

- $P(\text{Bin}(M, p^2 s^2) \geq 3) \leq O((M \cdot p^2 s^2)^3) = O(\frac{1}{n^{3-o(1)}})$
- $P(\text{Bin}(M, ps^2) \leq 2) \leq (1 - ps^2)^{M-2}(1 + (Mps^2) + \frac{(Mps^2)^2}{2})$

By Chernoff, we can guarantee $\lambda_\ell \geq (1 - \frac{1}{n^{1/3}})g(\rho_{\ell+1}\lambda_{\ell+1})$ whp.

Define $\ell_{thresh} = \lceil \frac{1}{3-\beta} \log_2(\frac{g^{-1}(0.6)}{C_3}) \rceil$. Then $\lambda_{\ell_{thresh}} \geq 0.5$.

Propagation III: use $L_{\ell+1}$ to partly match L_ℓ for $\ell \in [\ell_{min}, \ell_{thresh}]$.

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